

A rank-one counterexample to conjecture 00000001101

ziangni-sys

4 October 2026

Abstract

The root system A_1 disproves the claim that every root system has a Kazhdan–Lusztig polynomial whose leading coefficient has an odd prime factor. Its three nonzero Kazhdan–Lusztig polynomials are all the constant polynomial 1. A self-contained Lean 4 project proves the conclusion for every polynomial family satisfying the standard rank-one KL conditions, rather than merely checking a supplied table.

1 The statement and counterexample

The original English statement is:

Definition: The Kazhdan–Lusztig polynomials $P_{x,w}(q)$ of Weyl groups. Conjecture: For every root system there exist x, w such that the leading coefficient of $P_{x,w}$ contains an odd prime factor beyond powers of two (diversity of prime factors of KL coefficients).

The universal quantifier includes the reduced, irreducible, crystallographic root system $\Phi = \{-1, 1\} \subset \mathbb{R}$. The Euclidean inner product is $\langle a, b \rangle = ab$. For either root α , the root reflection is

$$r_\alpha(t) = t - 2 \frac{\langle t, \alpha \rangle}{\langle \alpha, \alpha \rangle} \alpha = -t.$$

Thus Φ spans its one-dimensional ambient space, is stable under root reflections, satisfies the reducedness condition, and its Cartan integers $2\langle \beta, \alpha \rangle / \langle \alpha, \alpha \rangle$ are ± 2 . This is the standard root system A_1 . Its Weyl group is $W = \{e, s\}$ with $s^2 = e$. The Coxeter lengths are $\ell(e) = 0$ and $\ell(s) = 1$; its Bruhat order is $e < s$.

2 Derivation of the KL polynomials

We use the usual convention $P_{x,w} = 0$ when $x \not\leq w$. For comparable pairs the KL polynomials obey

$$P_{w,w} = 1, \quad P_{x,w}(0) = 1, \quad \deg P_{x,w} \leq \frac{\ell(w) - \ell(x) - 1}{2} \quad (x < w).$$

For $e < s$, the degree bound is zero, so $P_{e,s}$ is constant; its constant term forces $P_{e,s} = 1$. Diagonal normalization gives the other two nonzero polynomials. Every pair in $W \times W$ is accounted for:

$P_{x,w}$	$w = e$	$w = s$
$x = e$	1	1
$x = s$	0	1

For completeness, the rank-one calculation can also be obtained directly from the defining Hecke-algebra construction, without using a precomputed table. Set $q = v^2$ and let $T = T_s$. The relation is

$$T^2 = (q - 1)T + q, \quad T^{-1} = q^{-1}T + (q^{-1} - 1).$$

The bar operation sends v to v^{-1} and T to T^{-1} . Consequently

$$\overline{v^{-1}(T + 1)} = v(q^{-1}T + (q^{-1} - 1) + 1) = v^{-1}(T + 1).$$

This element has the prescribed leading term $v^{-1}T$ and lower coefficient $v^{-1} \in v^{-1}\mathbb{Z}[v^{-1}]$. It is therefore the canonical basis element C'_s . Indeed, the difference of two such bar-invariant elements would be a scalar Laurent polynomial supported in strictly negative powers of v and equal to its bar, hence zero. Comparing

$$C'_s = v^{-\ell(s)} \sum_{x \leq s} P_{x,s}(v^2)T_x = v^{-1}(T + 1)$$

again yields $P_{e,s} = P_{s,s} = 1$.

Theorem 1. *Conjecture 00000001101 is false.*

Proof. Every nonzero KL polynomial of the root system A_1 equals 1, so its leading coefficient is 1. No prime divides 1: if a natural prime p divided 1, then $p = 1$, contradicting $p \geq 2$. In particular no odd prime divides the leading coefficient. The only remaining pair (s, e) has the zero polynomial, which has no leading nonzero coefficient and no prime factorization. It cannot witness the conjecture. The existential claim fails for this root system, disproving the universal statement. $\square$

3 Formal verification and semantic correspondence

The project uses Lean 4.19.0 and its standard library only. `W` is the two-element Weyl group indexing type, `action` is identity or integer negation, and `reflection_preserves_roots` checks preservation of the coordinate representatives $\{-1, 1\}$. `len` and `bruhat` encode the lengths and order above. `Poly` encodes an integer polynomial as a coefficient function $\mathbb{N} \rightarrow \mathbb{Z}$ together with a finite support bound.

`KLFamily P` is an explicit proposition on an arbitrary family of these polynomials. It records diagonal normalization, zero outside Bruhat order, constant term 1 for comparable pairs, and the strict degree bound coefficientwise: for $x < w$, a coefficient at k vanishes if $\ell(w) - \ell(x) \leq 2k$. This is exactly $2 \deg P_{x,w} < \ell(w) - \ell(x)$ for nonzero polynomials. In rank one these conditions uniquely determine all coefficients; `KL_unique_coeff` proves that fact. Thus the theorem applies to the actual KL family, which satisfies these standard conditions, and to any proposed implementation of it.

`IsLeading p k` requires the coefficient at k to be nonzero and all higher coefficients to vanish. This faithfully excludes the zero polynomial. `Prime` uses $p \geq 2$ and the usual positive natural divisor condition; `OddPrimeDivisor` requires primality, oddness and divisibility of the absolute integer coefficient. `leading_is_one` proves the leading coefficient is 1 for every pair and every proposed leading index. `rank_one_disproof` then proves the negation of the existential claim for every `KLFamily`; `certified_counterexample` instantiates it with a family whose defining conditions are themselves proved in `table_is_KL`.

No KL property is declared as a Lean axiom. The properties are explicit hypotheses of the generic theorem and are discharged for the rank-one witness. The project does not formalize general root-system or Hecke-algebra theory; the identification with A_1 and the direct canonical-basis derivation are provided above. No general KL positivity theorem or numerical search is needed. The final theorems depend only on the standard logical axioms `propext` and `Quot.sound`; there are no admitted proofs, custom axioms or native evaluation tactics.

4 Reproduction and sources

Run `lake build` and `lake env lean Main.lean` inside `lean/`. Run `python verify.py` in the submission directory for an independent exact Laurent-polynomial verification of the bar calculation and all four table entries. Compile this source with `tectonic report.tex`. The accompanying verification log records the compiler results and visual PDF inspection.

The mathematical conventions are those of D. Kazhdan and G. Lusztig, *Representations of Coxeter groups and Hecke algebras*, *Inventiones Mathematicae* 53 (1979), 165–184, doi:10.1007/BF01390031. For coefficient and indexing conventions see M. Goresky’s *Tables of Kazhdan–Lusztig Polynomials* (Institute for Advanced Study). The repository statement is `conjectures/00000001101.md`.